Session 1 — Linear algebra I

Solutions

These solutions are written to be read on their own, without the lecture. Each exercise begins with a shaded box recalling the one definition or formula it rests on and why that formula is what it is; then the computation is carried out with every intermediate step visible; then the result is read in words. The last line of many exercises names the mistake the question was built to catch — if you made it, you are in good company, and knowing which check would have caught it is worth more than the answer.

The table used throughout (scores out of 10). A *row* is one participant, a *column* is one measure:

	RT	acc.	anx.
P_1	4	9	1
P_2	6	7	3
P_3	5	8	2
P_4	9	4	6
P_5	7	6	4

Exercise 1 $\boldsymbol{a} = (1, 2)$, $\boldsymbol{b} = (3, -1)$ in $\mathbb{R}^2$; $\boldsymbol{u} = (1, -2, 4)$, $\boldsymbol{v} = (0, 3, -1)$ in $\mathbb{R}^3$. Compute sums and multiples; then solve $\lambda(1, 1) + \mu(1, -1) = (3, 5)$.

What this exercise uses. A vector of $\mathbb{R}^n$ is an ordered list of n numbers, and there are only *two* operations on it. **Addition** works slot by slot, $(a_1, \ldots, a_n) + (b_1, \ldots, b_n) = (a_1 + b_1, \ldots, a_n + b_n)$; **scaling** multiplies every slot by the same number, $\lambda(a_1, \ldots, a_n) = (\lambda a_1, \ldots, \lambda a_n)$. Nothing else is ever done to a vector in this course. One more fact is doing the real work in part (c): two vectors are *equal* exactly when all their coordinates agree, so one equation between vectors of $\mathbb{R}^2$ is really two ordinary equations.

Solution.

(a) Slot by slot:

$$\boldsymbol{a} + \boldsymbol{b} = (1 + 3,\ 2 + (-1)) = (4, 1), \qquad 2\boldsymbol{a} = (2 \cdot 1,\ 2 \cdot 2) = (2, 4),$$

$$\boldsymbol{a} - \boldsymbol{b} = \boldsymbol{a} + (-1)\boldsymbol{b} = (1 - 3,\ 2 - (-1)) = (-2, 3).$$

Note that $\boldsymbol{a} - \boldsymbol{b}$ is not a third operation: subtracting is adding $(-1)\boldsymbol{b}$.

(b) The same rule, now with three slots:

$$\boldsymbol{u} + 2\boldsymbol{v} = (1 + 2 \cdot 0,\ -2 + 2 \cdot 3,\ 4 + 2 \cdot (-1)) = (1, 4, 2),$$

$$3\boldsymbol{u} - \boldsymbol{v} = (3 \cdot 1 - 0,\ 3 \cdot (-2) - 3,\ 3 \cdot 4 - (-1)) = (3, -9, 13).$$

(c) First expand the left-hand side using the two operations:

$$\lambda(1, 1) + \mu(1, -1) = (\lambda, \lambda) + (\mu, -\mu) = (\lambda + \mu,\ \lambda - \mu).$$

Asking that this equal $(3,5)$ means asking that *each* coordinate match:

$$\begin{cases} \lambda + \mu = 3 & \text{(first coordinate)} \\ \lambda - \mu = 5 & \text{(second coordinate).} \end{cases}$$

Add the two lines: the μ cancels and $2\lambda = 8$, so $\lambda = 4$. Subtract the second from the first: the λ cancels and $2\mu = -2$, so $\mu = -1$. *Check — always do it:* $4(1,1) - 1 \cdot (1,-1) = (4,4) - (1,-1) = (3,5)$. ✓

Remark. Part (c) is the whole translation between geometry and algebra in miniature: one vector equation in $\mathbb{R}^n$ is n scalar equations. *Every question of the form "can this vector be built from those ones?" becomes a linear system this way, and that is what Exercises 4, 6 and 7 will do.*

Exercise 2 Rows and columns of the table: write P_1, P_2, P_3 as vectors, check $P_3 = \frac{1}{2}(P_1 + P_2)$, write the accuracy column, and say why a row and a column do not live in the same space.

What this exercise uses. A table of numbers can be read in two directions, and each reading produces vectors in a different space. Reading *across* a row collects the three measures of one person, so a row is a vector of $\mathbb{R}^3$. Reading *down* a column collects one measure over the five people, so a column is a vector of $\mathbb{R}^5$. The table is 5×3; the two numbers 5 and 3 are the dimensions of these two spaces.

Solution.

(a) Reading the first three rows across:

$$P_1 = (4,9,1), \qquad P_2 = (6,7,3), \qquad P_3 = (5,8,2).$$

Now compute the average of the first two — one addition, then one scaling:

$$P_1 + P_2 = (4+6,\ 9+7,\ 1+3) = (10,16,4), \qquad \tfrac{1}{2}(10,16,4) = (5,8,2).$$

That is exactly P_3. ✓ So participant 3 is, measure by measure, the midpoint of participants 1 and 2: an "average participant" of those two. Notice that this required nothing but the two operations of Exercise 1 — and that is all "combining participants" will ever mean.

(b) Reading the accuracy column downwards gives $(9,7,8,4,6)$, with one entry per participant. It is a vector of $\mathbb{R}^5$.

(c) Because a row has one coordinate *per measure* and a column has one coordinate *per participant*: $\mathbb{R}^3$ against $\mathbb{R}^5$. They answer two different questions — "what does this person look like?" and "how did this measure behave across people?" — and being lists of different lengths they cannot even be added to one another.

Common mistake. Adding a row to a column, or asking whether P_1 is "bigger" than the accuracy column. With a non-square table the mismatch is visible (3 numbers against 5); with a square table nothing warns you, so decide *in advance* which reading you are using. Software and papers switch between the two silently.

Exercise 3 Which of these pairs are linearly independent? Give a vanishing combination for each dependent one: (a) $(1,1),(2,2)$; (b) $(1,0),(0,1)$; (c) $(2,-4),(-1,2)$; (d) $(1,2,3),(2,4,7)$; (e) $(0,0),(1,2)$.

What this exercise uses. The definition: $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$ are **dependent** when some combination $\lambda_1 \boldsymbol{v}_1 + \cdots + \lambda_k \boldsymbol{v}_k = \mathbf{0}$ has coefficients *not all zero*, and **independent** otherwise. (Taking all $\lambda_i = 0$ always gives $\mathbf{0}$, which is why the definition insists on "not all zero".)

For exactly *two* vectors there is a shortcut: they are dependent iff one is a multiple of the other. Why: if $\lambda_1 \boldsymbol{v}_1 + \lambda_2 \boldsymbol{v}_2 = \mathbf{0}$ with, say, $\lambda_1 \neq 0$, then $\boldsymbol{v}_1 = -\frac{\lambda_2}{\lambda_1} \boldsymbol{v}_2$. Use the shortcut here — and only here: Exercise 4 shows it collapsing for three vectors.

Solution.

(a) **Dependent**: $(2,2) = 2(1,1)$. Rewritten as a vanishing combination, $2(1,1) - 1 \cdot (2,2) = (2,2) - (2,2) = \mathbf{0}$, with coefficients 2 and -1, not both zero.

(b) **Independent**: $\lambda(1,0) + \mu(0,1) = (\lambda, \mu)$, and this is $(0,0)$ only if $\lambda = 0$ *and* $\mu = 0$. No non-trivial combination vanishes. (This pair is the canonical basis of $\mathbb{R}^2$.)

(c) **Dependent**: $(-1,2) = -\frac{1}{2}(2,-4)$, i.e. $(2,-4) + 2(-1,2) = (2-2,\ -4+4) = \mathbf{0}$.

(d) **Independent**. Suppose $(2,4,7) = \lambda(1,2,3)$. The first coordinate forces $\lambda = 2$; the second is then consistent ($2 \cdot 2 = 4$); but the third would have to be $3 \cdot 2 = 6$, and it is 7. No λ works. Two vectors can agree on many coordinates and still fail to be proportional — one disagreement is enough.

(e) **Dependent**: $1 \cdot (0,0) + 0 \cdot (1,2) = \mathbf{0}$, and the coefficient 1 is not zero. *Any* family containing the zero vector is dependent, whatever the other vectors are, because $\mathbf{0}$ can always be given the nonzero coefficient.

Common mistake. Answering "independent" in (e) because $(0,0)$ and $(1,2)$ are "not multiples of each other". They are: $(0,0) = 0 \cdot (1,2)$. This case is the reason the official definition is the vanishing combination and not the "multiple of" phrasing.

Exercise 4 Is "no pair is proportional" a test of independence? (a) $(1,2,3)$, $(2,1,0)$, $(4,5,6)$; (b) $(1,1,0)$, $(0,1,1)$, $(1,3,2)$; (c) $(1,2)$ and $(3,5)$, which look nearly parallel; (d) what replaces the shortcut from three vectors on?

What this exercise uses. The shortcut of Exercise 3 answers the question "is one vector a multiple of *another single* vector?". Dependence asks something weaker and therefore easier to satisfy: "can one vector be built from *all the others together*?" Three vectors can be pairwise non-proportional and still have one of them sitting in the plane spanned by the other two. So from three vectors on, go back to the definition: look for $\lambda_1, \lambda_2, \lambda_3$, not all zero, with $\lambda_1 \boldsymbol{v}_1 + \lambda_2 \boldsymbol{v}_2 + \lambda_3 \boldsymbol{v}_3 = \mathbf{0}$. By Exercise 1 that vector equation is one ordinary equation per coordinate — a small *homogeneous* linear system, which you solve.

Solution.

(a) *No pair is proportional.* $\boldsymbol{v}_2 = \lambda \boldsymbol{v}_1$ would need $\lambda = 2$ (first coordinate) and $\lambda = \frac{1}{2}$ (second): impossible. $\boldsymbol{v}_3 = \lambda \boldsymbol{v}_1$ would need $\lambda = 4$ then $\lambda = \frac{5}{2}$: impossible. $\boldsymbol{v}_3 = \lambda \boldsymbol{v}_2$ would need $\lambda = 2$, and then the third coordinate would be $2 \cdot 0 = 0$, not 6: impossible.

And yet the family is ***dependent****.* Write the system: we want $\lambda_1(1,2,3) + \lambda_2(2,1,0) +$

$\lambda_3(4,5,6) = (0,0,0)$, i.e. one equation per coordinate,

$$\begin{cases} \lambda_1 + 2\lambda_2 + 4\lambda_3 = 0 & \text{(1st coordinate)} \\ 2\lambda_1 + \lambda_2 + 5\lambda_3 = 0 & \text{(2nd)} \\ 3\lambda_1 + 0 \cdot \lambda_2 + 6\lambda_3 = 0 & \text{(3rd).} \end{cases}$$

Start with the third line, the simplest: $3\lambda_1 = -6\lambda_3$, so

$$\boxed{\lambda_1 = -2\lambda_3}.$$

Put this into the first line: $-2\lambda_3 + 2\lambda_2 + 4\lambda_3 = 0$, i.e. $2\lambda_2 = -2\lambda_3$, so

$$\boxed{\lambda_2 = -\lambda_3}.$$

The second line must now be checked (it could have contradicted the others): $2(-2\lambda_3) + (-\lambda_3) + 5\lambda_3 = -4\lambda_3 - \lambda_3 + 5\lambda_3 = 0$ ✓— it is satisfied automatically, so there is no contradiction and λ_3 is free. Choosing $\lambda_3 = -1$ gives $\lambda_1 = 2$, $\lambda_2 = 1$, i.e.

$$2\boldsymbol{v}_1 + \boldsymbol{v}_2 - \boldsymbol{v}_3 = (2+2-4,\ 4+1-5,\ 6+0-6) = (0,0,0) = \mathbf{0}.\checkmark$$

A nonzero solution exists: **dependent**. (Any other choice of λ_3 gives a multiple of the same relation — the solutions form a line.)

(b) Again no pair is proportional: each pair has a coordinate where one vanishes and the other does not, or a ratio that is not constant — e.g. $(1,3,2) = \lambda(1,1,0)$ would need $\lambda = 1$ then $\lambda = 3$. Yet

$$(1,1,0) + 2(0,1,1) - (1,3,2) = (1+0-1,\ 1+2-3,\ 0+2-2) = \mathbf{0}.$$

Dependent. Geometrically: $(1,1,0)$ and $(0,1,1)$ span a plane, and $(1,3,2)$ already lies in that plane, so it adds no new direction. (Compare Exercise 7(b), where $(1,2,1) = (1,1,0) + (0,1,1)$ lies in the same plane.)

(c) **Independent**, appearances notwithstanding. Suppose $\lambda(1,2) + \mu(3,5) = \mathbf{0}$; coordinate by coordinate,

$$\begin{cases} \lambda + 3\mu = 0 \\ 2\lambda + 5\mu = 0. \end{cases}$$

The first line gives $\lambda = -3\mu$; substituting into the second, $2(-3\mu) + 5\mu = -6\mu + 5\mu = -\mu = 0$, so $\mu = 0$, and then $\lambda = 0$. Only the trivial combination vanishes. (Equivalent one-line check, as in Exercise 5: $\det\left(\begin{smallmatrix} 1 & 3 \\ 2 & 5 \end{smallmatrix}\right) = 1 \cdot 5 - 3 \cdot 2 = -1 \neq 0$.) Two vectors at a small angle are still independent: what matters is not how close the directions look, but whether they are *exactly* the same.

(d) From three vectors on, the only reliable test is the definition itself: solve the homogeneous system $\sum_i \lambda_i \boldsymbol{v}_i = \mathbf{0}$ and ask whether a nonzero solution exists. Next session this gets a name and a systematic algorithm — computing the *rank*.

Common mistake. Checking the vectors two at a time. Pairwise proportionality is a property of pairs; dependence is a property of the whole family. (a) and (b) are exactly the case where the two answers differ — and note that the reverse error exists too, (c): the eye compares how the lists *look*, the definition compares what can be *built*.

Exercise 5 For which values of t are $(t, 2)$ and $(8, t)$ linearly dependent?

What this exercise uses. Two vectors of $\mathbb{R}^2$, written as the columns of a 2×2 matrix, are dependent exactly when the determinant of that matrix is 0: $\det\left(\begin{smallmatrix} a & c \\ b & d \end{smallmatrix}\right) = ad - bc$. Exercise 25 explains *why* — the determinant is the area of the parallelogram built on the two vectors, and that area collapses to 0 precisely when the two vectors lie on one line.

Solution. Put the two vectors in the columns and set the determinant to zero:

$$\det\begin{pmatrix} t & 8 \\ 2 & t \end{pmatrix} = t \cdot t - 8 \cdot 2 = t^2 - 16 = 0 \iff t^2 = 16 \iff t = 4 \text{ or } t = -4.$$

Check both: for $t = 4$ the vectors are $(4, 2)$ and $(8, 4) = 2(4, 2)$ ✓; for $t = -4$ they are $(-4, 2)$ and $(8, -4) = -2(-4, 2)$ ✓.

The same thing without determinants, if you prefer: dependence means $(8, t) = \lambda(t, 2)$ for some λ. The second coordinate gives $t = 2\lambda$; the first gives $8 = \lambda t = \lambda(2\lambda) = 2\lambda^2$, so $\lambda^2 = 4$, $\lambda = \pm 2$, and $t = 2\lambda = \pm 4$. Same answer, and it shows what the determinant condition is really encoding.

Common mistake. Forgetting $t = -4$. A quadratic equation has two roots; "$t^2 = 16$" is not "$t = 4$".

Exercise 6 Are $(1, 2)$, $(2, 4)$, $(0, 1)$ independent in $\mathbb{R}^2$? Answer first by a dimension count, then explicitly; and can a 3-item questionnaire produce more than 3 independent columns?

What this exercise uses. A counting fact, and it is the most economical tool in the whole session: in $\mathbb{R}^n$, **any** $n + 1$ **vectors are dependent**. The reason is visible in the system: k vectors of $\mathbb{R}^n$ give n equations in k unknowns $\lambda_1, \ldots, \lambda_k$, all of them homogeneous (right-hand side $\mathbf{0}$). With more unknowns than equations such a system always has a solution besides the all-zero one. So the answer can be given before any computation.

Solution.

(a) **No**, and nothing needs computing: these are 3 vectors in $\mathbb{R}^2$, a space of dimension 2, and $3 > 2$. In system terms: 3 unknowns, 2 equations, hence a nonzero solution exists.

(b) An explicit one. Spot that $(2, 4) = 2(1, 2)$; the third vector can then be given the coefficient 0:

$$2(1, 2) - 1 \cdot (2, 4) + 0 \cdot (0, 1) = (2 - 2,\ 4 - 4) = \mathbf{0},$$

with coefficients $(2, -1, 0)$ — not all zero, as required.

(c) **No.** The table has exactly 3 columns, so at most 3 of them can be independent, however many people answer. This is worth getting right because the intuition pulls the other way: adding respondents makes each column *longer* (a vector in a bigger $\mathbb{R}^n$), it does not make the columns *more numerous*. Length and number are different things, and it is the number that is capped.

Remark. This counting fact is what becomes "$\text{rank} \le \min(\text{rows}, \text{columns})$*" next session — so a* 5×3 *table can never have more than 3 independent directions of variation.*

Exercise 7 (a) Is $(2,5)$ in $\text{span}\{(1,1),(1,-1)\}$? (b) Describe $\text{span}\{(1,1,0),(0,1,1)\}$ in $\mathbb{R}^3$; is $(1,2,1)$ in it? (c) Why does every span contain $\mathbf{0}$?

What this exercise uses. $\text{span}(\boldsymbol{v}_1,\dots,\boldsymbol{v}_k)$ is the set of *all* linear combinations $\lambda_1\boldsymbol{v}_1+\cdots+\lambda_k\boldsymbol{v}_k$, the λ_i running over all real numbers. It is what those vectors can build, and nothing outside it is reachable however clever the coefficients. Testing "is $\boldsymbol{x}$ in the span?" is therefore testing whether a linear system has a solution — the same translation as Exercise 1(c). The *shape* of a span: k independent vectors span a k-dimensional flat — a line for 1, a plane for 2 — always passing through the origin.

Solution.

(a) Look for λ,μ with $\lambda(1,1)+\mu(1,-1)=(2,5)$, i.e.

$$\begin{cases}\lambda+\mu=2\\ \lambda-\mu=5.\end{cases}$$

Adding: $2\lambda=7$, $\lambda=\frac{7}{2}$. Subtracting: $2\mu=-3$, $\mu=-\frac{3}{2}$. A solution exists, so **yes**:

$$(2,5)=\tfrac{7}{2}(1,1)-\tfrac{3}{2}(1,-1).$$

Check: $\left(\frac{7}{2}-\frac{3}{2},\ \frac{7}{2}+\frac{3}{2}\right)=(2,5)$. ✓ In fact $(1,1)$ and $(1,-1)$ are independent, so by Exercise 6's counting fact they span all of $\mathbb{R}^2$: *every* vector of the plane is reachable, and the answer had to be yes.

(b) The two vectors are independent (neither is a multiple of the other: the first coordinate would force $\lambda=0$, the second $\lambda=1$). Two independent vectors span a 2-dimensional flat through the origin, i.e. a **plane through the origin** sitting inside $\mathbb{R}^3$. Is $(1,2,1)$ on it? Solve $\lambda(1,1,0)+\mu(0,1,1)=(1,2,1)$: the first coordinate gives $\lambda=1$, the third gives $\mu=1$, and the middle one must then agree — $1+1=2$ ✓. So **yes**, with coefficients 1 and 1:

$$(1,1,0)+(0,1,1)=(1,2,1).$$

(c) Take every coefficient equal to 0: $0\boldsymbol{v}_1+\cdots+0\boldsymbol{v}_k=\mathbf{0}$, which is a legitimate linear combination. So $\mathbf{0}$ belongs to every span. This is why a span is never an arbitrary blob of space: it is a line, a plane or a higher flat *through the origin.*

Remark. Practical consequence for data: a cloud of measurements that does not surround the origin is not a span. That is one reason one centres *the data — subtracts the mean of each column — before speaking of the directions along which it varies.*

Exercise 8 Dot products, norms, and the unit vector along $(5,-12)$.

What this exercise uses. The **dot product** multiplies the two lists slot by slot and adds up: $\boldsymbol{a}\cdot\boldsymbol{b}=a_1b_1+\cdots+a_nb_n$. It returns *one number*, not a vector. Taking the dot product of a vector with itself gives the sum of the squares of its coordinates, which by Pythagoras is the square of its length — hence the **norm** $\|\boldsymbol{a}\|=\sqrt{\boldsymbol{a}\cdot\boldsymbol{a}}$. A **unit vector** is one of norm 1; to build the unit vector pointing along $\boldsymbol{a}$, divide $\boldsymbol{a}$ by its own norm, which changes the length to 1 and leaves the direction untouched.

Solution.

(a) Multiply coordinate by coordinate, then add:

$$(1,2,3)\cdot(4,0,-1) = 1\cdot 4 + 2\cdot 0 + 3\cdot(-1) = 4 + 0 - 3 = 1,$$

$$(2,-3)\cdot(4,1) = 8 - 3 = 5, \qquad (1,0,-2)\cdot(3,5,1) = 3 + 0 - 2 = 1.$$

(b) Sum of squares, then square root:

$$\|(3,4)\| = \sqrt{3^2+4^2} = \sqrt{25} = 5, \qquad \|(6,8)\| = \sqrt{36+64} = \sqrt{100} = 10,$$

$$\|(1,1,1)\| = \sqrt{1+1+1} = \sqrt{3} \approx 1.732, \qquad \|(2,-2,1)\| = \sqrt{4+4+1} = \sqrt{9} = 3.$$

Observe $(6,8) = 2(3,4)$ and $10 = 2\times 5$: scaling a vector by λ scales its length by $|\lambda|$.

(c) $\|(5,-12)\| = \sqrt{25+144} = \sqrt{169} = 13$, so the unit vector is

$$\frac{(5,-12)}{13} = \left(\tfrac{5}{13}, -\tfrac{12}{13}\right).$$

Check: $\left(\tfrac{5}{13}\right)^2 + \left(\tfrac{12}{13}\right)^2 = \tfrac{25+144}{169} = 1.$ ✓

Common mistake. Writing a vector where the dot product gives a number. $(1,2,3)\cdot(4,0,-1)$ is 1, not $(4,0,-3)$: the multiplication slot by slot is only the first half of the recipe, the sum is the second.

Exercise 9 Weights $\boldsymbol{w} = (1,2,-1)$ — reward speed a little, accuracy a lot, penalise anxiety. Score the five participants; who is highest and lowest; what happens with $2\boldsymbol{w}$; and find a nonzero profile scoring 0.

What this exercise uses. A **composite score** — a scale total, an index, the prediction of a linear model — takes several measures and returns one number by weighting and adding. That is precisely a dot product: $\text{score}(\boldsymbol{p}) = \boldsymbol{w}\cdot\boldsymbol{p}$, where $\boldsymbol{w}$ holds the weights and $\boldsymbol{p}$ one participant's profile. Being a dot product, it is *linear* in each argument, and the two facts used below are consequences: $(\lambda\boldsymbol{w})\cdot\boldsymbol{p} = \lambda(\boldsymbol{w}\cdot\boldsymbol{p})$, and score 0 means exactly $\boldsymbol{w}\perp\boldsymbol{p}$.

Solution.

(a) Take each row of the table and dot it with $\boldsymbol{w} = (1,2,-1)$:

$$\begin{aligned}
P_1 = (4,9,1):&\quad 1(4)+2(9)-1(1) &&= 4+18-1 = 21,\\
P_2 = (6,7,3):&\quad 1(6)+2(7)-1(3) &&= 6+14-3 = 17,\\
P_3 = (5,8,2):&\quad 1(5)+2(8)-1(2) &&= 5+16-2 = 19,\\
P_4 = (9,4,6):&\quad 1(9)+2(4)-1(6) &&= 9+8-6 = 11,\\
P_5 = (7,6,4):&\quad 1(7)+2(6)-1(4) &&= 7+12-4 = 15.
\end{aligned}$$

(b) Highest: P_1 with 21; lowest: P_4 with 11. This is what the weights were built to do — P_1 is accurate and calm, P_4 is the reverse — so the ranking is a sanity check on the rule, not a discovery about the data.

(c) Since the score is linear in the weights, doubling them doubles every score:

$$42, \qquad 34, \qquad 38, \qquad 22, \qquad 30.$$

What changed: the numbers. **What did not:** the ranking of the participants, and every ratio between scores. The rules $\boldsymbol{w}$ and $2\boldsymbol{w}$ say the same thing — "accuracy counts twice as much as speed" — in a different unit. So a raw score has no meaning on its own and two rules cannot be compared through their raw scores. What *can* be reported is the score measured with a unit ruler,

$$\frac{\boldsymbol{w} \cdot \boldsymbol{p}}{\|\boldsymbol{w}\|},$$

which is unchanged if $\boldsymbol{w}$ is rescaled: replacing $\boldsymbol{w}$ by $2\boldsymbol{w}$ multiplies numerator and denominator both by 2. That quantity is the *shadow* of Exercises 10 and 13.

(d) We need $\boldsymbol{p} \neq \boldsymbol{0}$ with $\boldsymbol{w} \cdot \boldsymbol{p} = 0$, i.e. $p_1 + 2p_2 - p_3 = 0$: one equation, three unknowns, so there are many solutions. Pick values freely for two coordinates and solve for the third: with $p_1 = 1, p_2 = 0$ we get $p_3 = 1$, giving $\boldsymbol{p} = (1, 0, 1)$ (check: $1+0-1 = 0$ ✓); with $p_1 = 2, p_2 = -1$ we get $p_3 = 0$, giving $(2, -1, 0)$ (check: $2-2+0 = 0$ ✓). The set of all such profiles is a *plane* through the origin — two free coordinates, hence dimension 2. Exercise 10 explains why this plane is the blind spot of the rule.

Exercise 10 $\boldsymbol{w} = (1, 2, -1)$, $\boldsymbol{p} = (1, 1, 0)$, $\boldsymbol{p}' = (3, 0, 0)$: compute both scores; explain via $\boldsymbol{p}' - \boldsymbol{p}$; compute the shadow on $\boldsymbol{w}/\|\boldsymbol{w}\|$; what does it mean for a study?

What this exercise uses. Two readings of the same number, and this exercise is about the second. *Computationally*, $\boldsymbol{w} \cdot \boldsymbol{p} = \sum_i w_i p_i$. *Geometrically*,

$$\boldsymbol{w} \cdot \boldsymbol{p} = \|\boldsymbol{w}\| \times \|\boldsymbol{p}\| \times \cos\alpha,$$

where α is the angle between the two vectors. Three factors: $\|\boldsymbol{w}\|$ is an arbitrary unit chosen with the weights (Exercise 9(c)), $\|\boldsymbol{p}\|$ is the participant's overall level, and only $\cos\alpha$ says anything about the *shape* of the profile. In particular, if $\alpha = 90°$ then $\cos\alpha = 0$ and the score is 0 regardless of how large $\boldsymbol{p}$ is.

Solution.

(a)

$$\boldsymbol{w} \cdot \boldsymbol{p} = 1(1) + 2(1) - 1(0) = 3, \qquad \boldsymbol{w} \cdot \boldsymbol{p}' = 1(3) + 2(0) - 1(0) = 3.$$

The same score, 3 — and the two participants are not remotely alike. $\boldsymbol{p}$ is average on speed and on accuracy; $\boldsymbol{p}'$ is fast and gets *nothing* right. Whatever the difference between these two people is, this score does not see it.

(b)

$$\boldsymbol{p}' - \boldsymbol{p} = (3-1,\ 0-1,\ 0-0) = (2, -1, 0), \qquad \boldsymbol{w} \cdot (2, -1, 0) = 2 - 2 + 0 = 0.$$

This explains (a) with no coincidence left in it. The dot product is linear, so

$$\boldsymbol{w} \cdot \boldsymbol{p}' = \boldsymbol{w} \cdot (\boldsymbol{p} + (\boldsymbol{p}' - \boldsymbol{p})) = \underbrace{\boldsymbol{w} \cdot \boldsymbol{p}}_{=3} + \underbrace{\boldsymbol{w} \cdot (\boldsymbol{p}' - \boldsymbol{p})}_{=0} = 3.$$

The two profiles differ by a direction *orthogonal* to $\boldsymbol{w}$, and moving along such a direction costs nothing in score. More generally, the set of profiles with a given score is the whole plane perpendicular to $\boldsymbol{w}$ passing through any one of them; inside that plane you may move as far as you like and the score will not budge.

(c) $\|\boldsymbol{w}\| = \sqrt{1^2 + 2^2 + (-1)^2} = \sqrt{6} \approx 2.449$, so

$$\boldsymbol{p} \cdot \frac{\boldsymbol{w}}{\|\boldsymbol{w}\|} = \frac{\boldsymbol{w} \cdot \boldsymbol{p}}{\|\boldsymbol{w}\|} = \frac{3}{\sqrt{6}} \approx 1.22. \checkmark$$

It is the same shadow, read with a ruler of length 1 instead of a ruler of length $\|\boldsymbol{w}\|$ — which is why it is the version that survives rescaling the weights. (And $\boldsymbol{p}'$ gives 1.22 too: same shadow, which is exactly what (b) said.)

(d) That the study is structurally blind in that direction. Participants may differ enormously along $(2, -1, 0)$ and the analysis will report *nothing* — not a weak effect, nothing — while the difference is perfectly real. The decision that caused this was taken when the weights were chosen, so it is a design problem, not a statistical one, and no amount of extra data will fix it.

Remark. This is also why two questionnaires can correlate at zero and still both be measuring something real: they may simply be looking along perpendicular directions.

Exercise 11 (a) Are $(1, 2)$ and $(2, -1)$ orthogonal, and what does that mean for a scoring rule $\boldsymbol{w} = (1, 2)$ applied to $\boldsymbol{p} = (2, -1)$? (b) For which k are $(k, 1)$ and $(k, -4)$ orthogonal?

What this exercise uses. Orthogonal (perpendicular) means $\boldsymbol{a} \cdot \boldsymbol{b} = 0$. It follows from the geometric formula $\boldsymbol{a} \cdot \boldsymbol{b} = \|\boldsymbol{a}\| \, \|\boldsymbol{b}\| \cos \alpha$: for nonzero vectors the product vanishes exactly when $\cos \alpha = 0$, i.e. $\alpha = 90°$. The sign of the dot product likewise reads off the angle — positive below $90°$, negative beyond.

Solution.

(a) $(1, 2) \cdot (2, -1) = 1 \cdot 2 + 2 \cdot (-1) = 2 - 2 = 0$: **yes, orthogonal**. For the scoring rule this is not a curiosity but a verdict: the profile $\boldsymbol{p} = (2, -1)$ receives the score 0. Someone can be strongly "$(2, -1)$-shaped", and twice or ten times as strongly so — since $\boldsymbol{w} \cdot (10\boldsymbol{p}) = 10 \times 0 = 0$ — and the rule will still return zero. It is blind to that whole direction, exactly as in Exercise 10.

(b) Compute the dot product with k kept symbolic and set it to zero:

$$(k, 1) \cdot (k, -4) = k \cdot k + 1 \cdot (-4) = k^2 - 4 = 0 \quad \Longleftrightarrow \quad k = 2 \ \text{ or } \ k = -2.$$

Check $k = 2$: $(2, 1) \cdot (2, -4) = 4 - 4 = 0$ ✓. *Check* $k = -2$: $(-2, 1) \cdot (-2, -4) = 4 - 4 = 0$ ✓.

Exercise 12 (a) $\cos \alpha$ for $(1, 2)$ and $(3, -1)$; (b) are $(1, 2)$ and $(2, 1)$ perpendicular; (c) cosine similarity of the profiles $(2, 4)$ and $(4, 8)$, and what it ignores.

What this exercise uses. Rearranging $\boldsymbol{a} \cdot \boldsymbol{b} = \|\boldsymbol{a}\| \, \|\boldsymbol{b}\| \cos \alpha$ gives a formula for the angle from the coordinates alone:

$$\cos \alpha = \frac{\boldsymbol{a} \cdot \boldsymbol{b}}{\|\boldsymbol{a}\| \, \|\boldsymbol{b}\|}.$$

Dividing by the two lengths removes any information about *size* and keeps only *direction*; the result always lies in $[-1, 1]$ (that is Cauchy–Schwarz, Exercise 16). Applied to two data profiles it is called the **cosine similarity**, and it is how two word vectors are compared in a semantic-space model, or two participants irrespective of their overall level.

Solution.

(a) $\boldsymbol{a} \cdot \boldsymbol{b} = 1 \cdot 3 + 2 \cdot (-1) = 1$, $\|\boldsymbol{a}\| = \sqrt{1+4} = \sqrt{5}$, $\|\boldsymbol{b}\| = \sqrt{9+1} = \sqrt{10}$, so

$$\cos\alpha = \frac{1}{\sqrt{5}\,\sqrt{10}} = \frac{1}{\sqrt{50}} = \frac{1}{5\sqrt{2}} = \frac{\sqrt{2}}{10} \approx 0.141 \qquad (\alpha \approx 81.9^\circ).$$

Nearly perpendicular, but not quite: the dot product is small, not zero.

(b) $(1,2) \cdot (2,1) = 2 + 2 = 4 \neq 0$, so **not** perpendicular. Both norms are $\sqrt{5}$, hence

$$\cos\alpha = \frac{4}{\sqrt{5} \cdot \sqrt{5}} = \frac{4}{5} = 0.8 \qquad (\alpha \approx 36.9^\circ).$$

(c) $(4,8) = 2(2,4)$: the second profile is the first, doubled. So the angle is 0 and

$$\cos\alpha = \frac{2 \cdot 4 + 4 \cdot 8}{\sqrt{4+16}\,\sqrt{16+64}} = \frac{8+32}{\sqrt{20}\sqrt{80}} = \frac{40}{\sqrt{1600}} = \frac{40}{40} = 1.$$

Meaning: the two participants have exactly the same *shape* of profile — the same balance between the two measures. *What it ignores:* the overall level. The second participant scores twice as high on everything, and the cosine cannot see that at all. Whether this is a feature or a bug depends on your question: it is a feature if you want to compare profiles of people who differ in overall engagement or scale use, and a bug if the level is precisely what you are studying.

Exercise 13 Project onto *unit* directions: (a) $(3,1)$ onto $\boldsymbol{u} = \frac{1}{\sqrt{2}}(1,1)$; (b) $(1,3)$ onto $\frac{1}{\sqrt{2}}(1,-1)$; (c) the residual in (a), and its orthogonality to $\boldsymbol{u}$.

What this exercise uses. Shine a light perpendicular to a line and look at the shadow a vector casts on it. For a **unit** vector $\boldsymbol{u}$ the two ingredients are:

$$\text{signed shadow length} = \boldsymbol{a} \cdot \boldsymbol{u}, \qquad \text{projection vector} = (\boldsymbol{a} \cdot \boldsymbol{u})\,\boldsymbol{u}.$$

The first is a *number* (it can be negative, meaning the shadow falls on the other side); the second is a *vector* on the line. The identity behind them is $\boldsymbol{a} \cdot \boldsymbol{u} = \|\boldsymbol{a}\|\,\|\boldsymbol{u}\| \cos\alpha = \|\boldsymbol{a}\| \cos\alpha$ since $\|\boldsymbol{u}\| = 1$ — "adjacent over hypotenuse" times the hypotenuse. What is left over, $\boldsymbol{a} - (\boldsymbol{a} \cdot \boldsymbol{u})\boldsymbol{u}$, is called the **residual**: the part of $\boldsymbol{a}$ the direction $\boldsymbol{u}$ cannot account for.

Solution.

(a) $\boldsymbol{a} = (3,1)$, $\boldsymbol{u} = \frac{1}{\sqrt{2}}(1,1)$. First the number:

$$\boldsymbol{a} \cdot \boldsymbol{u} = \frac{3 \cdot 1 + 1 \cdot 1}{\sqrt{2}} = \frac{4}{\sqrt{2}} = \frac{4}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = 2\sqrt{2} \approx 2.828.$$

Then the vector:

$$(\boldsymbol{a} \cdot \boldsymbol{u})\,\boldsymbol{u} = 2\sqrt{2} \cdot \tfrac{1}{\sqrt{2}}(1,1) = 2(1,1) = (2,2).$$

(b) $(1,3)$ onto $\frac{1}{\sqrt{2}}(1,-1)$:

$$\boldsymbol{a} \cdot \boldsymbol{u} = \frac{1 \cdot 1 + 3 \cdot (-1)}{\sqrt{2}} = \frac{-2}{\sqrt{2}} = -\sqrt{2} \approx -1.414,$$

$$(\boldsymbol{a} \cdot \boldsymbol{u})\,\boldsymbol{u} = -\sqrt{2} \cdot \tfrac{1}{\sqrt{2}}(1,-1) = -(1,-1) = (-1,1).$$

The shadow length is *negative*: the projection falls on the side of the line opposite to $\boldsymbol{u}$. That sign is information, not an error.

(c) Residual $= \boldsymbol{a} - (\boldsymbol{a}\cdot\boldsymbol{u})\boldsymbol{u} = (3,1) - (2,2) = (1,-1)$, and

$$(1,-1)\cdot\tfrac{1}{\sqrt{2}}(1,1) = \frac{1-1}{\sqrt{2}} = 0.\checkmark$$

So the vector splits as

$$\underbrace{(3,1)}_{\boldsymbol{a}} = \underbrace{(2,2)}_{\text{what the axis sees}} + \underbrace{(1,-1)}_{\text{what it misses}},$$

and the two pieces are perpendicular. That split is the picture behind "explained + unexplained" in every linear model you will ever fit.

Common mistake. Answering with the number only. "Project" asks for both: the shadow *length* $2\sqrt{2}$ and the projection *vector* $(2,2)$ are different objects living in different places.

Exercise 14 $\boldsymbol{a} = (3,1)$ projected onto directions that are *not* unit vectors: (a) onto $\boldsymbol{v} = (1,1)$; (b) onto $(2,2)$ and $(-1,-1)$; (c) is the projection the closest point of the line? (d) project the projection again.

What this exercise uses. For a direction $\boldsymbol{v}$ of *any* length the formula is

$$P_{\boldsymbol{v}}(\boldsymbol{a}) = \frac{\boldsymbol{a}\cdot\boldsymbol{v}}{\boldsymbol{v}\cdot\boldsymbol{v}}\,\boldsymbol{v}.$$

Where does the denominator come from? Write the unit vector as $\boldsymbol{u} = \boldsymbol{v}/\left\|\boldsymbol{v}\right\|$ and substitute into the formula of Exercise 13:

$$(\boldsymbol{a}\cdot\boldsymbol{u})\boldsymbol{u} = \left(\boldsymbol{a}\cdot\frac{\boldsymbol{v}}{\|\boldsymbol{v}\|}\right)\frac{\boldsymbol{v}}{\|\boldsymbol{v}\|} = \frac{\boldsymbol{a}\cdot\boldsymbol{v}}{\|\boldsymbol{v}\|^2}\,\boldsymbol{v} = \frac{\boldsymbol{a}\cdot\boldsymbol{v}}{\boldsymbol{v}\cdot\boldsymbol{v}}\,\boldsymbol{v}.$$

So the division by $\boldsymbol{v}\cdot\boldsymbol{v} = \|\boldsymbol{v}\|^2$ is exactly the normalisation you would otherwise have done by hand; if $\|\boldsymbol{v}\| = 1$ it equals 1 and disappears.

Solution.

(a) $\boldsymbol{a}\cdot\boldsymbol{v} = 3+1 = 4$ and $\boldsymbol{v}\cdot\boldsymbol{v} = 1+1 = 2$, so

$$P_{\boldsymbol{v}}(\boldsymbol{a}) = \frac{4}{2}\,(1,1) = 2(1,1) = (2,2).$$

This is the **same projection vector** as in Exercise 13(a). What differs is the *coordinate* describing it: the multiplier is 2 when the ruler is $\boldsymbol{v}$ itself, and $2\sqrt{2}$ when the ruler is the unit vector $\boldsymbol{u}$. One shadow, two ways of measuring it — and the unit one is what belongs in a table, because it does not depend on an arbitrary choice.

(b) Onto $(2,2)$: $\dfrac{6+2}{4+4}(2,2) = \dfrac{8}{8}(2,2) = (2,2)$.

Onto $(-1,-1)$: $\dfrac{-3-1}{1+1}(-1,-1) = -2\,(-1,-1) = (2,2)$.

Always $(2,2)$. The projection depends only on the *line* — not on which vector was picked to name it, not on its length, not even on which way it points (in the last case both the coefficient and the vector changed sign, and the two changes cancelled). This is precisely what the denominator buys.

(c) The residual is $\boldsymbol{a} - P_{\boldsymbol{v}}(\boldsymbol{a}) = (3,1) - (2,2) = (1,-1)$, so the distance from $\boldsymbol{a}$ to its projection is

$$\|(1,-1)\| = \sqrt{1+1} = \sqrt{2} \approx 1.414,$$

while the distance to the point $(1,1)$ of the same line is

$$\|(3,1)-(1,1)\| = \|(2,0)\| = 2.$$

Since $\sqrt{2} < 2$, the projection is the better of the two — and in fact it beats *every* point of the line. Here is why, in one computation. Take any point $t\boldsymbol{v}$ of the line and split

$$\boldsymbol{a} - t\boldsymbol{v} = \underbrace{\big(\boldsymbol{a} - P_{\boldsymbol{v}}(\boldsymbol{a})\big)}_{\perp\text{ the line}} + \underbrace{\big(P_{\boldsymbol{v}}(\boldsymbol{a}) - t\boldsymbol{v}\big)}_{\text{along the line}}.$$

The two pieces are perpendicular, so Pythagoras applies:

$$\|\boldsymbol{a} - t\boldsymbol{v}\|^2 = \|\boldsymbol{a} - P_{\boldsymbol{v}}(\boldsymbol{a})\|^2 + \|P_{\boldsymbol{v}}(\boldsymbol{a}) - t\boldsymbol{v}\|^2 \;\geq\; \|\boldsymbol{a} - P_{\boldsymbol{v}}(\boldsymbol{a})\|^2,$$

with equality only when the second term vanishes, i.e. $t\boldsymbol{v} = P_{\boldsymbol{v}}(\boldsymbol{a})$. **The projection is the closest point of the line** — and this inequality *is* least squares: fitting a linear model means projecting the data onto the span of the predictors, and "minimising the sum of squared residuals" means choosing that closest point.

(d)

$$P_{\boldsymbol{v}}((2,2)) = \frac{2+2}{2}(1,1) = 2(1,1) = (2,2):$$

nothing happens. It was predictable: $(2,2)$ already lies on the line, and by (c) the closest point of the line to a point already on it is that point itself. In symbols $P \circ P = P$, or $\boldsymbol{P}^2 = \boldsymbol{P}$ once the projection is written as a matrix — the shadow of a shadow is the same shadow.

Common mistake. Using the unit-vector formula $(\boldsymbol{a}\cdot\boldsymbol{v})\boldsymbol{v}$ with a $\boldsymbol{v}$ that is not unit. Here it would give $4(1,1) = (4,4)$ instead of $(2,2)$ — a vector twice too long, and no longer the closest point of the line.

Exercise 15 $\boldsymbol{e}_1 = \frac{1}{\sqrt{2}}(1,1,0)$, $\boldsymbol{e}_2 = (0,0,1)$, $S = \operatorname{span}\{\boldsymbol{e}_1, \boldsymbol{e}_2\}$, $\boldsymbol{a} = (2,4,5)$: check orthonormality, compute $P_S(\boldsymbol{a})$, check the residual.

What this exercise uses. To project onto a plane (or any subspace S), project onto each axis of an **orthonormal** basis of S and add the results:

$$P_S(\boldsymbol{a}) = (\boldsymbol{a}\cdot\boldsymbol{e}_1)\,\boldsymbol{e}_1 + \cdots + (\boldsymbol{a}\cdot\boldsymbol{e}_k)\,\boldsymbol{e}_k.$$

So a plane is no harder than a line: it is one line-projection per basis vector. This works *only* because the $\boldsymbol{e}_i$ are orthonormal — each axis measures a piece of $\boldsymbol{a}$ that the others do not see, so the terms can simply be added without interfering. With skewed axes the sum would double-count and the formula would be wrong.

Solution.

(a) Norms: $\|\boldsymbol{e}_1\|^2 = \frac{1}{2}(1^2+1^2+0^2) = \frac{1}{2}\cdot 2 = 1$ and $\|\boldsymbol{e}_2\|^2 = 0+0+1 = 1$, so both are unit vectors. Orthogonality: $\boldsymbol{e}_1\cdot\boldsymbol{e}_2 = \frac{1}{\sqrt{2}}(1\cdot 0 + 1\cdot 0 + 0\cdot 1) = 0$. ✓ The family is orthonormal, so the formula above may be used.

(b) The two coefficients first:

$$\boldsymbol{a}\cdot\boldsymbol{e}_1 = \frac{2\cdot 1 + 4\cdot 1 + 5\cdot 0}{\sqrt{2}} = \frac{6}{\sqrt{2}} = 3\sqrt{2}, \qquad \boldsymbol{a}\cdot\boldsymbol{e}_2 = 2\cdot 0 + 4\cdot 0 + 5\cdot 1 = 5.$$

Then the two projection vectors, and their sum:

$$P_S(\boldsymbol{a}) = 3\sqrt{2} \cdot \tfrac{1}{\sqrt{2}}(1,1,0) + 5\,(0,0,1) = 3(1,1,0) + (0,0,5) = (3,3,0) + (0,0,5) = (3,3,5).$$

(c) Residual:

$$\boldsymbol{a} - P_S(\boldsymbol{a}) = (2,4,5) - (3,3,5) = (-1,1,0),$$

$$(-1,1,0)\cdot \boldsymbol{e}_1 = \frac{-1+1+0}{\sqrt{2}} = 0, \qquad (-1,1,0)\cdot \boldsymbol{e}_2 = 0.\checkmark$$

And that is enough to conclude for *every* vector of S: any such vector is $\lambda \boldsymbol{e}_1 + \mu \boldsymbol{e}_2$, so by linearity its dot product with the residual is $\lambda \cdot 0 + \mu \cdot 0 = 0$. Checking the two basis vectors checks the whole plane.

Remark. Compare the shapes: $(2,4,5) \mapsto (3,3,5)$ *has left the third coordinate untouched (that direction is* in S*) and replaced the first two by their common average* 3 *(the plane only records the sum of the first two coordinates, not how it is split between them).*

Exercise 16 Check Cauchy–Schwarz for $\boldsymbol{a} = (1,2)$, $\boldsymbol{b} = (2,4)$: is it an equality, and why? Then: why does $(10,0)\cdot(10,0) = 100$ not contradict "the dot product lies between -1 and 1"?

What this exercise uses. Cauchy–Schwarz: $|\boldsymbol{a}\cdot\boldsymbol{b}| \le \|\boldsymbol{a}\|\,\|\boldsymbol{b}\|$ always, with equality exactly when the two vectors are dependent (collinear). It is what makes the angle well defined: dividing the inequality by $\|\boldsymbol{a}\|\,\|\boldsymbol{b}\|$ gives $\left|\frac{\boldsymbol{a}\cdot\boldsymbol{b}}{\|\boldsymbol{a}\|\|\boldsymbol{b}\|}\right| \le 1$, so that quotient is legitimately the cosine of something. Without the inequality, $\cos\alpha$ could come out at 1.7 and mean nothing.

Solution. Compute the two sides separately.

$$\boldsymbol{a}\cdot\boldsymbol{b} = 1\cdot 2 + 2\cdot 4 = 2 + 8 = 10, \qquad \text{so} \quad |\boldsymbol{a}\cdot\boldsymbol{b}| = 10;$$

$$\|\boldsymbol{a}\| = \sqrt{1+4} = \sqrt{5}, \quad \|\boldsymbol{b}\| = \sqrt{4+16} = \sqrt{20}, \qquad \|\boldsymbol{a}\|\,\|\boldsymbol{b}\| = \sqrt{5}\cdot\sqrt{20} = \sqrt{100} = 10.$$

The two sides are equal: this is the **equality case**. The reason is visible in the data — $\boldsymbol{b} = 2\boldsymbol{a}$, so the vectors are collinear, the angle between them is 0 and $\cos\alpha = 1$; the inequality $|\cos\alpha| \le 1$ is saturated. Cauchy–Schwarz is an equality precisely in this situation and strict otherwise.

The second question. There is no contradiction because the dot product is *not* the quantity bounded by 1. Here $(10,0)\cdot(10,0) = 100$, and Cauchy–Schwarz says only that this is at most $\|(10,0)\|^2 = 10 \times 10 = 100$ — which it is, with equality (a vector is collinear with itself). What always lies in $[-1,1]$ is the **normalised** product

$$\cos\alpha = \frac{\boldsymbol{a}\cdot\boldsymbol{b}}{\|\boldsymbol{a}\|\,\|\boldsymbol{b}\|} \qquad \Big(\text{here } \frac{100}{100} = 1\Big).$$

Common mistake. "The dot product is between -1 and 1." That sentence quietly assumes both vectors are unit vectors — true in that case, false in general. It is the cosine, not the dot product, that is bounded.

Exercise 17 $\boldsymbol{w} = (1,1)$ and unit profiles $\boldsymbol{p} = (\cos\theta, \sin\theta)$: which gives the largest score, which give zero, and what does a zero-scoring direction mean?

What this exercise uses. Restricting to profiles of *length one* removes the "size" factor from $\boldsymbol{w} \cdot \boldsymbol{p} = \|\boldsymbol{w}\| \, \|\boldsymbol{p}\| \cos\alpha$ and leaves $\boldsymbol{w} \cdot \boldsymbol{p} = \|\boldsymbol{w}\| \cos\alpha$: among profiles of equal size, the score *is* an angle. Parametrising the unit circle as $\boldsymbol{p} = (\cos\theta, \sin\theta)$ turns the question into an ordinary one-variable maximisation.

Solution. Compute the score as a function of θ:

$$\boldsymbol{w} \cdot \boldsymbol{p} = 1 \cdot \cos\theta + 1 \cdot \sin\theta = \cos\theta + \sin\theta.$$

Rewrite it as a single cosine — factor out $\sqrt{2}$ so that the two coefficients become a cosine and a sine:

$$\cos\theta + \sin\theta = \sqrt{2}\Big(\tfrac{1}{\sqrt{2}}\cos\theta + \tfrac{1}{\sqrt{2}}\sin\theta\Big) = \sqrt{2}(\cos 45^\circ \cos\theta + \sin 45^\circ \sin\theta) = \sqrt{2}\,\cos(\theta - 45^\circ),$$

using $\cos(x - y) = \cos x \cos y + \sin x \sin y$. Everything follows from this one line.

(a) The cosine is largest, equal to 1, when its argument is 0, i.e. $\theta = 45^\circ$. The winning profile is

$$\boldsymbol{p} = (\cos 45^\circ, \sin 45^\circ) = \tfrac{1}{\sqrt{2}}(1,1) = \frac{\boldsymbol{w}}{\|\boldsymbol{w}\|},$$

the unit vector *pointing the same way as the weights*, and the score is $\sqrt{2} = \|\boldsymbol{w}\|$. So a scoring rule is looking for its own direction: the best possible profile, at fixed size, is a copy of $\boldsymbol{w}$.

(b) The score is 0 when $\cos(\theta - 45^\circ) = 0$, i.e. $\theta - 45^\circ = \pm 90^\circ$, giving $\theta = 135^\circ$ and $\theta = 315^\circ$. These are the two unit vectors $\pm\frac{1}{\sqrt{2}}(-1,1)$ — exactly the directions orthogonal to $\boldsymbol{w}$ (check: $(1,1) \cdot (-1,1) = 0$ ✓).

(c) That participants can differ substantially along that direction while the score records none of it. The instrument is not weakly sensitive there, it is blind; and since the blindness was fixed the moment the weights were chosen, it is a design problem rather than a rounding error. Exercise 10 is this same statement in $\mathbb{R}^3$ with two named participants instead of an angle.

Exercise 18 (a) Is $\{(1,0),(0,1)\}$ orthonormal? And $\{(1,1),(1,-1)\}$? (b) Make the second orthonormal. (c) Complete $\frac{1}{\sqrt{2}}(1,-1)$ into an orthonormal basis of $\mathbb{R}^2$. (d) An orthonormal basis whose first axis points along $(3,4)$.

What this exercise uses. Orthonormal = *ortho* (the vectors are mutually perpendicular: all pairwise dot products are 0) + *normal* (each has norm 1). Two separate conditions, and a family can satisfy one without the other. Orthonormality is what makes coordinates cheap: in such a basis the coordinates of a vector are just dot products (Exercise 19), and lengths are computed by Pythagoras.

Solution.

(a) $\{(1,0),(0,1)\}$: dot product $1 \cdot 0 + 0 \cdot 1 = 0$, and both have norm 1 — **yes**, orthonormal (it is the canonical basis).
$\{(1,1),(1,-1)\}$: dot product $1 \cdot 1 + 1 \cdot (-1) = 0$, so they *are* perpendicular; but $\|(1,1)\| = \sqrt{2} \neq 1$. So the family is **orthogonal but not orthonormal**. Only the second condition fails.

(b) Failing only normality is easy to repair — divide each vector by its own norm (which does not change any angle, so orthogonality survives):

$$\left\{\tfrac{1}{\sqrt{2}}(1,1),\ \tfrac{1}{\sqrt{2}}(1,-1)\right\}.$$

(c) We need a unit vector orthogonal to $\frac{1}{\sqrt{2}}(1,-1)$. In $\mathbb{R}^2$ there is a recipe: rotating by 90° sends (a,b) to $(-b,a)$, so $(1,-1)\mapsto(1,1)$. Normalising, a valid completion is

$$\left\{\tfrac{1}{\sqrt{2}}(1,-1),\ \tfrac{1}{\sqrt{2}}(1,1)\right\}$$

(the second vector could equally be its opposite $-\frac{1}{\sqrt{2}}(1,1)$ — the answer is not unique).

(d) Normalise $(3,4)$, whose norm is $\sqrt{9+16}=5$: the first axis is $\frac{1}{5}(3,4)$. Apply the same 90° recipe to get the second: $(3,4)\mapsto(-4,3)$, of norm 5 as well, so

$$\left\{\tfrac{1}{5}(3,4),\ \tfrac{1}{5}(-4,3)\right\}.$$

Check: $(3,4)\cdot(-4,3)=-12+12=0$ ✓, and both have norm 1 after division by 5 ✓.

Remark. The recipe $(a,b)\mapsto(-b,a)$ always works in the plane, because $(a,b)\cdot(-b,a)=-ab+ab=0$ whatever a and b are. In $\mathbb{R}^3$ and beyond there is no such single formula — you use Gram–Schmidt instead (Exercise 23).

Exercise 19 In the orthonormal basis $\boldsymbol{e}_1=\frac{1}{5}(3,4)$, $\boldsymbol{e}_2=\frac{1}{5}(-4,3)$: the coordinates of $(1,0)$; the coordinates of $(5,5)$ with a rebuild check; and the length check.

What this exercise uses. Coordinates in an orthonormal basis are dot products: $x_k=\boldsymbol{x}\cdot\boldsymbol{e}_k$. Why: writing $\boldsymbol{x}=x_1\boldsymbol{e}_1+x_2\boldsymbol{e}_2$ and dotting both sides with $\boldsymbol{e}_1$ gives $\boldsymbol{x}\cdot\boldsymbol{e}_1=x_1(\boldsymbol{e}_1\cdot\boldsymbol{e}_1)+x_2(\boldsymbol{e}_2\cdot\boldsymbol{e}_1)=x_1\cdot 1+x_2\cdot 0=x_1$. No system to solve — that is exactly what orthonormality buys.

Two checks are then compulsory. **Rebuild**: recompute $x_1\boldsymbol{e}_1+x_2\boldsymbol{e}_2$ and see whether the original vector comes back. **Lengths**: $x_1^2+x_2^2$ must equal $\|\boldsymbol{x}\|^2$ (Pythagoras, valid because the axes are perpendicular and of length 1).

Solution.

(a) For $\boldsymbol{x}=(1,0)$:

$$x_1=(1,0)\cdot\tfrac{1}{5}(3,4)=\tfrac{3}{5},\qquad x_2=(1,0)\cdot\tfrac{1}{5}(-4,3)=-\tfrac{4}{5}.$$

So in the tilted axes the old first basis vector reads $\left(\frac{3}{5},-\frac{4}{5}\right)$ — the vector has not moved, only its description has.

(b) For $\boldsymbol{x}=(5,5)$:

$$x_1=\frac{5\cdot 3+5\cdot 4}{5}=\frac{15+20}{5}=7,\qquad x_2=\frac{5\cdot(-4)+5\cdot 3}{5}=\frac{-20+15}{5}=-1.$$

Rebuild check:

$$7\cdot\tfrac{1}{5}(3,4)-1\cdot\tfrac{1}{5}(-4,3)=\left(\tfrac{21}{5},\tfrac{28}{5}\right)-\left(-\tfrac{4}{5},\tfrac{3}{5}\right)=\left(\tfrac{21+4}{5},\ \tfrac{28-3}{5}\right)=\left(\tfrac{25}{5},\tfrac{25}{5}\right)=(5,5).✓$$

(c) *Length check:* $x_1^2 + x_2^2 = 49 + 1 = 50$, and $\|(5,5)\|^2 = 25 + 25 = 50$. ✓ The two checks are independent and catch different errors — Exercise 21(d) shows one error that only the first detects and one that only the second detects, which is why you run both.

Exercise 20 The contrast basis $\boldsymbol{f}_1 = \frac{1}{\sqrt{2}}(1,1)$ ("overall level"), $\boldsymbol{f}_2 = \frac{1}{\sqrt{2}}(-1,1)$ ("B − A"): coordinates of $(2,0)$ with a rebuild; coordinates of $(1,3)$ with an interpretation; the length check.

What this exercise uses. Same machinery as Exercise 19 — the basis is orthonormal (check: $(1,1)\cdot(-1,1) = -1+1 = 0$, and each has norm 1 after the $\frac{1}{\sqrt{2}}$), so coordinates are dot products. What is new is the *reading.* Think of two similar tasks A and B: the raw coordinates are the two scores, and the new ones are "how good overall" (along $\boldsymbol{f}_1$) and "how much better at B than at A" (along $\boldsymbol{f}_2$). Recoding a pair of scores as level and difference is nothing but this change of basis — and it is exactly what a *contrast* does in an experimental design.

Solution.

(a) $\boldsymbol{x} = (2,0)$:

$$x_1 = \boldsymbol{x}\cdot\boldsymbol{f}_1 = \frac{2\cdot 1 + 0\cdot 1}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2}, \qquad x_2 = \boldsymbol{x}\cdot\boldsymbol{f}_2 = \frac{2\cdot(-1) + 0\cdot 1}{\sqrt{2}} = -\sqrt{2}.$$

Rebuild check: $\sqrt{2}\,\boldsymbol{f}_1 - \sqrt{2}\,\boldsymbol{f}_2 = (1,1) - (-1,1) = (2,0)$. ✓

(b) $\boldsymbol{x} = (1,3)$:

$$x_1 = \frac{1+3}{\sqrt{2}} = \frac{4}{\sqrt{2}} = 2\sqrt{2} \approx 2.83, \qquad x_2 = \frac{-1+3}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2} \approx 1.41.$$

In words: the first number is large and positive — a high **overall level** across the two tasks. The second is positive too, so this participant is **better at task B than at task A**. The pair (level, difference) carries exactly the same information as the pair (score A, score B): nothing was lost or gained, the same point is being described from better-chosen axes.

(c) $x_1^2 + x_2^2 = (2\sqrt{2})^2 + (\sqrt{2})^2 = 8 + 2 = 10$, and $\|(1,3)\|^2 = 1 + 9 = 10$. ✓ Lengths are preserved because the new axes are perpendicular and of length 1 — this is Pythagoras, and it is precisely what would fail with skewed axes.

Exercise 21 $\boldsymbol{B}$ has columns $\boldsymbol{f}_1 = \frac{1}{\sqrt{2}}(1,1)$, $\boldsymbol{f}_2 = \frac{1}{\sqrt{2}}(-1,1)$. (a) Write $\boldsymbol{B}$ and check $\boldsymbol{B}^\top\boldsymbol{B} = \boldsymbol{I}$; (b) compute $\boldsymbol{B}^\top\boldsymbol{x}$ for $\boldsymbol{x} = (3,1)$; (c) compute $\boldsymbol{B}\boldsymbol{x}_{\text{new}}$; (d) which check exposes which of two wrong answers?

What this exercise uses. The change of basis of Exercises 19–20 can be packaged into one matrix. Collect the new basis vectors as the **columns** of $\boldsymbol{B}$. Then:

$$\underbrace{\boldsymbol{x}_{\text{new}} = \boldsymbol{B}^\top\boldsymbol{x}}_{\text{out: measure}}, \qquad \underbrace{\boldsymbol{x} = \boldsymbol{B}\,\boldsymbol{x}_{\text{new}}}_{\text{back: rebuild}}.$$

Both are matrix–vector products, but they are read differently. Transposing turns the columns of $\boldsymbol{B}$ into the *rows* of $\boldsymbol{B}^\top$, and each entry of a matrix–vector product is (row)·(vector) — so $\boldsymbol{B}^\top\boldsymbol{x}$ computes the dot products $\boldsymbol{f}_i\cdot\boldsymbol{x}$, one measurement per axis. Read the other way, $\boldsymbol{B}\boldsymbol{x}_{\text{new}}$ is a combination of the *columns* with the coordinates as weights: $x_1\boldsymbol{f}_1 + x_2\boldsymbol{f}_2$, a reconstruction.

Solution.

(a) Columns are the basis vectors, so

$$\boldsymbol{B} = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & -1\\ 1 & 1\end{pmatrix}, \qquad \boldsymbol{B}^\top = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ -1 & 1\end{pmatrix}$$

(transposing swaps rows and columns). Then

$$\boldsymbol{B}^\top\boldsymbol{B} = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ -1 & 1\end{pmatrix}\cdot\frac{1}{\sqrt{2}}\begin{pmatrix}1 & -1\\ 1 & 1\end{pmatrix} = \frac{1}{2}\begin{pmatrix}1+1 & -1+1\\ -1+1 & 1+1\end{pmatrix} = \frac{1}{2}\begin{pmatrix}2 & 0\\ 0 & 2\end{pmatrix} = \boldsymbol{I}.\checkmark$$

What the identity says: entry (i,j) of $\boldsymbol{B}^\top\boldsymbol{B}$ is $\boldsymbol{f}_i\cdot\boldsymbol{f}_j$, so "$\boldsymbol{B}^\top\boldsymbol{B} = \boldsymbol{I}$" is the single equation "the columns are unit vectors (1 on the diagonal) and mutually orthogonal (0 off it)" — orthonormality in one line. *What it saves:* it says $\boldsymbol{B}^{-1} = \boldsymbol{B}^\top$, so the way back is a transpose, which costs nothing. **No matrix is ever inverted for an orthonormal change of basis.**

(b)

$$\boldsymbol{x}_{\text{new}} = \boldsymbol{B}^\top\boldsymbol{x} = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ -1 & 1\end{pmatrix}\begin{pmatrix}3\\ 1\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}3+1\\ -3+1\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}4\\ -2\end{pmatrix} = \begin{pmatrix}2\sqrt{2}\\ -\sqrt{2}\end{pmatrix}.$$

The two dot products give the same: $\boldsymbol{x}\cdot\boldsymbol{f}_1 = \frac{3+1}{\sqrt{2}} = 2\sqrt{2}$ and $\boldsymbol{x}\cdot\boldsymbol{f}_2 = \frac{-3+1}{\sqrt{2}} = -\sqrt{2}$. ✓ They must, since row i of $\boldsymbol{B}^\top$ *is* $\boldsymbol{f}_i$.

(c)

$$\boldsymbol{B}\,\boldsymbol{x}_{\text{new}} = 2\sqrt{2}\,\boldsymbol{f}_1 - \sqrt{2}\,\boldsymbol{f}_2 = 2\sqrt{2}\cdot\tfrac{1}{\sqrt{2}}(1,1) - \sqrt{2}\cdot\tfrac{1}{\sqrt{2}}(-1,1) = (2,2)-(-1,1) = (3,1) = \boldsymbol{x}.\checkmark$$

In words: this product **rebuilds** the arrow. It takes the coordinates measured in the new axes and returns the description of the same arrow in the canonical axes $(1,0),(0,1)$, where it reads $(3,1)$. Nothing moved; only the language changed, and this is the trip back.

(d) Both errors are caught without redoing the computation — but not by the same check.

- $(3\sqrt{2}, -2\sqrt{2})$: the **length check** kills it, since $(3\sqrt{2})^2 + (2\sqrt{2})^2 = 18+8 = 26$, whereas $\|(3,1)\|^2 = 9+1 = 10$. (What went wrong: the projections were never taken — the coordinates were scaled by $\sqrt{2}$ rather than measured against the axes.)
- $\sqrt{2}\,(\boldsymbol{f}_1 - 2\boldsymbol{f}_2)$, i.e. $(\sqrt{2}, -2\sqrt{2})$: here $2+8=10$, so the length check *passes* and is useless. It is **rebuilding** that exposes it:

$$\sqrt{2}\,\boldsymbol{f}_1 - 2\sqrt{2}\,\boldsymbol{f}_2 = (1,1)+(2,-2) = (3,-1) \neq (3,1).$$

  (What went wrong: the two coefficients were swapped — the large one went to the second axis.)

Remark. This is exactly why both checks are compulsory rather than optional. The length check is blind to any permutation or sign change of the coordinates — those preserve $x_1^2 + x_2^2$ — and rebuilding is the one that sees them. Between them they catch essentially every slip of this kind.

Exercise 22 (a) Axes rotated by 30°: write $\boldsymbol{e}_1, \boldsymbol{e}_2$ and the coordinates of $(2,0)$. (b) Verify $Q^\top Q = I$ for $Q = \begin{pmatrix}3/5 & -4/5\\ 4/5 & 3/5\end{pmatrix}$. (c) Why does an orthonormal change of basis preserve lengths?

What this exercise uses. A unit vector at angle θ from the horizontal is $(\cos\theta, \sin\theta)$. The second axis is the first rotated by a further $90°$, and rotating by $90°$ sends (a, b) to $(-b, a)$, hence $\boldsymbol{e}_2 = (-\sin\theta, \cos\theta)$. A matrix whose columns are orthonormal is called **orthogonal** and satisfies $Q^\mathsf{T}Q = I$; rotations are the standard example.

Solution.

(a) With $\theta = 30°$, using $\cos 30° = \frac{\sqrt{3}}{2}$ and $\sin 30° = \frac{1}{2}$:

$$\boldsymbol{e}_1 = \left(\tfrac{\sqrt{3}}{2},\ \tfrac{1}{2}\right), \qquad \boldsymbol{e}_2 = \left(-\tfrac{1}{2},\ \tfrac{\sqrt{3}}{2}\right).$$

The basis is orthonormal, so the coordinates of $(2, 0)$ are dot products:

$$x_1 = (2,0)\cdot\boldsymbol{e}_1 = 2\cdot\tfrac{\sqrt{3}}{2} + 0 = \sqrt{3} \approx 1.73, \qquad x_2 = (2,0)\cdot\boldsymbol{e}_2 = 2\cdot\left(-\tfrac{1}{2}\right) + 0 = -1.$$

Length check: $x_1^2 + x_2^2 = 3 + 1 = 4 = \|(2,0)\|^2$. ✓ The second coordinate is negative because the vector lies clockwise of the new second axis.

(b) Entry (i, j) of $Q^\mathsf{T}Q$ is (column i of Q)·(column j of Q). The columns are $\frac{1}{5}(3,4)$ and $\frac{1}{5}(-4,3)$, so

$$(1,1):\ \tfrac{9}{25} + \tfrac{16}{25} = 1, \qquad (2,2):\ \tfrac{16}{25} + \tfrac{9}{25} = 1, \qquad (1,2) = (2,1):\ \tfrac{-12}{25} + \tfrac{12}{25} = 0.$$

Hence $Q^\mathsf{T}Q = \left(\begin{smallmatrix}1&0\\0&1\end{smallmatrix}\right) = I$: the columns are orthonormal and Q is an orthogonal matrix. (These are the very axes of Exercise 19.)

(c) Two ways to see it. *Geometrically:* the coordinates are the projections onto perpendicular axes of length 1, so the vector is the hypotenuse of a right triangle with legs x_1 and x_2, and Pythagoras gives $\|\boldsymbol{x}\|^2 = x_1^2 + x_2^2$ in *any* orthonormal basis — the total does not depend on the axes. *Algebraically:*

$$\|Q\boldsymbol{x}\|^2 = (Q\boldsymbol{x})^\mathsf{T}(Q\boldsymbol{x}) = \boldsymbol{x}^\mathsf{T}Q^\mathsf{T}Q\,\boldsymbol{x} = \boldsymbol{x}^\mathsf{T}I\,\boldsymbol{x} = \boldsymbol{x}^\mathsf{T}\boldsymbol{x} = \|\boldsymbol{x}\|^2,$$

where the whole argument rests on the $Q^\mathsf{T}Q = I$ just verified.

Exercise 23 From $(1,1)$ and $(1,0)$, build an orthonormal basis of $\mathbb{R}^2$: first vector as the first axis, subtract from the second its component along that axis, normalise.

What this exercise uses. This is **Gram–Schmidt**, and it is nothing but Exercise 13(c) applied twice: keep the first direction, and from the second vector keep only the *residual* — the part the first axis cannot see, which is orthogonal to it by construction. Then normalise each. The procedure works in any dimension, which is why it exists at all (in $\mathbb{R}^2$ the $90°$ trick of Exercise 18 is quicker).

Solution. **Step 1 — first axis.** Normalise $(1,1)$, whose norm is $\sqrt{2}$:

$$\boldsymbol{f}_1 = \frac{(1,1)}{\sqrt{2}} = \tfrac{1}{\sqrt{2}}(1,1).$$

Step 2 — remove from $\boldsymbol{a} = (1,0)$ its component along $\boldsymbol{f}_1$.

$$\boldsymbol{a}\cdot\boldsymbol{f}_1 = \frac{1\cdot 1 + 0\cdot 1}{\sqrt{2}} = \frac{1}{\sqrt{2}}, \qquad (\boldsymbol{a}\cdot\boldsymbol{f}_1)\,\boldsymbol{f}_1 = \frac{1}{\sqrt{2}}\cdot\tfrac{1}{\sqrt{2}}(1,1) = \tfrac{1}{2}(1,1) = \left(\tfrac{1}{2}, \tfrac{1}{2}\right),$$

$$\boldsymbol{r} = \boldsymbol{a} - (\boldsymbol{a} \cdot \boldsymbol{f}_1)\boldsymbol{f}_1 = (1,0) - \left(\tfrac{1}{2}, \tfrac{1}{2}\right) = \left(\tfrac{1}{2}, -\tfrac{1}{2}\right).$$

Step 3 — normalise the residual. $\|\boldsymbol{r}\| = \sqrt{\frac{1}{4} + \frac{1}{4}} = \frac{1}{\sqrt{2}}$, so

$$\boldsymbol{f}_2 = \frac{\boldsymbol{r}}{\|\boldsymbol{r}\|} = \sqrt{2}\left(\tfrac{1}{2}, -\tfrac{1}{2}\right) = \tfrac{1}{\sqrt{2}}(1,-1).$$

Checks: $\boldsymbol{f}_1 \cdot \boldsymbol{f}_2 = \frac{1\cdot 1 + 1\cdot(-1)}{2} = 0$ ✓, and both have norm 1 ✓. The orthonormal basis is $\{\frac{1}{\sqrt{2}}(1,1),\ \frac{1}{\sqrt{2}}(1,-1)\}$.

Common mistake. Normalising at the wrong moment — for instance normalising $(1,0)$ first and then subtracting. Subtract first, normalise last: it is the *residual* that must end up with length 1, and its direction is not the direction of $(1,0)$.

Exercise 24 Compute and describe in words: (a) $\left(\begin{smallmatrix}2&0\\0&3\end{smallmatrix}\right)\left(\begin{smallmatrix}1\\1\end{smallmatrix}\right)$; (b) $\left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)\left(\begin{smallmatrix}5\\-2\end{smallmatrix}\right)$; (c) which matrix rotates the plane by 90° anticlockwise?

What this exercise uses. A matrix is a *machine*: it eats a vector and returns a vector, moving every point of space at once. To compute $\boldsymbol{A}\boldsymbol{x}$, entry i of the output is (row i of $\boldsymbol{A}$)$\cdot\boldsymbol{x}$:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} ax_1 + bx_2 \\ cx_1 + dx_2 \end{pmatrix}.$$

And to *write down* a matrix from a geometric description, use the fact that column j is the image of the j-th basis vector (Exercise 25 makes this explicit): decide where $(1,0)$ and $(0,1)$ must go, and put those two answers in the columns.

Solution.

(a)

$$\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2\cdot 1 + 0\cdot 1 \\ 0\cdot 1 + 3\cdot 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}.$$

A *diagonal* matrix scales each axis independently: it stretches horizontally by 2 and vertically by 3. (Data reading: converting two measures into different units.)

(b)

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 5 \\ -2 \end{pmatrix} = \begin{pmatrix} 0\cdot 5 + 1\cdot(-2) \\ 1\cdot 5 + 0\cdot(-2) \end{pmatrix} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}.$$

It swaps the two coordinates, which geometrically is the reflection in the line $y = x$ (points on that line, where the coordinates are already equal, do not move).

(c) Use the column rule. A 90° anticlockwise rotation sends $(1,0) \mapsto (0,1)$ and $(0,1) \mapsto (-1,0)$; put these in the columns:

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Check on $(1,0)$*:* $R\left(\begin{smallmatrix}1\\0\end{smallmatrix}\right) = \left(\begin{smallmatrix}0\cdot 1 + (-1)\cdot 0\\ 1\cdot 1 + 0\cdot 0\end{smallmatrix}\right) = \left(\begin{smallmatrix}0\\1\end{smallmatrix}\right)$. ✓ (This is the $(a,b) \mapsto (-b,a)$ recipe of Exercise 18, now written as a matrix.)

Exercise 25 $\boldsymbol{A} = \left(\begin{smallmatrix}2&1\\1&2\end{smallmatrix}\right)$: (a) images of the basis vectors; (b) $\boldsymbol{A}(1,1)$ and $\boldsymbol{A}(2,1)$ without multiplying; (c) the area factor; (d) the same for $\boldsymbol{C} = \left(\begin{smallmatrix}1&2\\2&4\end{smallmatrix}\right)$; (e) the sign of $\det\left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$.

What this exercise uses. Two facts, both about *columns*.

1. Column j is the image of the j-th basis vector. Indeed $\boldsymbol{A}\binom{1}{0}$ picks out $1 \times (\text{column } 1) + 0 \times (\text{column } 2)$. More generally $\boldsymbol{Ax} = x_1\boldsymbol{c}_1 + x_2\boldsymbol{c}_2$: the output is a linear combination of the columns, with the entries of $\boldsymbol{x}$ as coefficients. So the set of all possible outputs is exactly span(columns).

2. $|\det \boldsymbol{A}|$ is the factor by which areas are multiplied. The unit square, spanned by $(1,0)$ and $(0,1)$, is sent to the parallelogram built on the two columns, whose area is $|\det \boldsymbol{A}| = |ad - bc|$. Since the plane is tiled by copies of that square and a linear map treats every copy the same way, *every* region is scaled by that same number. The sign says whether the plane was flipped over.

Solution.

(a)

$$\boldsymbol{A}\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}2\\1\end{pmatrix}, \qquad \boldsymbol{A}\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}1\\2\end{pmatrix}.$$

These are read straight off the matrix: they are its first and second columns. No arithmetic is required, and that is the point of the title.

(b) Write $\boldsymbol{c}_1 = (2,1)$, $\boldsymbol{c}_2 = (1,2)$ and use $\boldsymbol{Ax} = x_1\boldsymbol{c}_1 + x_2\boldsymbol{c}_2$:

$$\boldsymbol{A}(1,1) = \boldsymbol{c}_1 + \boldsymbol{c}_2 = (2+1,\ 1+2) = (3,3),$$

$$\boldsymbol{A}(2,1) = 2\boldsymbol{c}_1 + \boldsymbol{c}_2 = (4,2) + (1,2) = (5,4).$$

Reading a matrix–vector product as a combination of columns is usually faster than the row recipe, and always more informative.

(c) The unit square has area 1; its image is the parallelogram on $\boldsymbol{c}_1$ and $\boldsymbol{c}_2$, of area

$$|\det \boldsymbol{A}| = |2 \cdot 2 - 1 \cdot 1| = 3.$$

And *every* region — a disc, a triangle, a blob — has its area multiplied by the same 3. Why the same: the plane is tiled by copies of the unit square, a linear map sends each copy to a congruent copy of the same parallelogram, and any shape can be approximated as finely as you like by such squares. One number therefore summarises the effect on area everywhere, and that number is the determinant. Here it is positive, so orientation is preserved.

(d) For $\boldsymbol{C} = \left(\begin{smallmatrix}1&2\\2&4\end{smallmatrix}\right)$:

$$\boldsymbol{C}\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}1\\2\end{pmatrix}, \qquad \boldsymbol{C}\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}2\\4\end{pmatrix} = 2\begin{pmatrix}1\\2\end{pmatrix},$$

$$\boldsymbol{C}(1,1) = \begin{pmatrix}1\\2\end{pmatrix} + \begin{pmatrix}2\\4\end{pmatrix} = \begin{pmatrix}3\\6\end{pmatrix}, \qquad \boldsymbol{C}(2,1) = 2\begin{pmatrix}1\\2\end{pmatrix} + \begin{pmatrix}2\\4\end{pmatrix} = \begin{pmatrix}2\\4\end{pmatrix} + \begin{pmatrix}2\\4\end{pmatrix} = \begin{pmatrix}4\\8\end{pmatrix}.$$

The two columns are collinear, so the parallelogram is flat and

$$\det \boldsymbol{C} = 1 \cdot 4 - 2 \cdot 2 = 0.$$

The unit square is flattened onto a segment, of area 0. The set of all outputs is

$$\{\boldsymbol{Cx} \ : \ \boldsymbol{x} \in \mathbb{R}^2\} = \operatorname{span}\{(1,2)\},$$

a *line*, not the plane — and indeed the four images computed above, $(1,2)$, $(2,4)$, $(3,6)$, $(4,8)$, all lie on it. Two different inputs can now share an image: $\boldsymbol{C}(2,0) = (2,4)$ and $\boldsymbol{C}(0,1) = (2,4)$. Information is destroyed, so there is no way back and $\boldsymbol{C}$ is not invertible.

(e)

$$\det\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = 0 \cdot 0 - 1 \cdot 1 = -1.$$

Its *absolute value* is 1: areas are unchanged. Its *sign* is negative: the orientation of the plane is reversed. Consistently with Exercise 24(b), this matrix is a reflection in the line $y = x$, and a mirror preserves areas while exchanging left and right — unlike a rotation, whose determinant is $+1$.

Remark. Two things to carry into Session 2: $\det = 0$ *means "flattened, information lost, not invertible" (it will be the statement that the rank is not full), and for this* $\boldsymbol{A}$ *the eigenvalues turn out to be* 1 *and* 3*, whose product* $1 \times 3 = 3 = \det \boldsymbol{A}$ *is not a coincidence.*

Exercise 26 $\boldsymbol{M} = \begin{pmatrix} 1/2 & 1/2 \\ -1 & 1 \end{pmatrix}$, $\boldsymbol{s} = (8,4)$: compute $\boldsymbol{M}\boldsymbol{s}$ by rows and by columns; read the output; is $\boldsymbol{M}$ an orthonormal change of basis; is $\boldsymbol{M}^\top$ the way back?

What this exercise uses. The same product read two ways. *By rows*: entry i of the output is (row i)$\cdot\boldsymbol{s}$ — this is how you compute. *By columns*: $\boldsymbol{M}\boldsymbol{s} = s_1\boldsymbol{c}_1 + s_2\boldsymbol{c}_2$ — this is how you understand. They always agree. And a caution carried over from Exercise 21: $\boldsymbol{B}^\top\boldsymbol{B} = \boldsymbol{I}$ requires the columns to be orthogonal *and* of norm 1. Orthogonality alone is not enough, and when it fails the transpose is not the inverse.

Solution.

(a) *By rows* — each output entry is a dot product of a row with $\boldsymbol{s}$:

$$\boldsymbol{M}\boldsymbol{s} = \begin{pmatrix} (\frac{1}{2}, \frac{1}{2}) \cdot (8,4) \\ (-1,1) \cdot (8,4) \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \cdot 8 + \frac{1}{2} \cdot 4 \\ -8 + 4 \end{pmatrix} = \begin{pmatrix} 4 + 2 \\ -4 \end{pmatrix} = \begin{pmatrix} 6 \\ -4 \end{pmatrix}.$$

By columns — a combination of the columns with 8 and 4 as weights:

$$8\begin{pmatrix} \frac{1}{2} \\ -1 \end{pmatrix} + 4\begin{pmatrix} \frac{1}{2} \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ -8 \end{pmatrix} + \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 6 \\ -4 \end{pmatrix}. \checkmark$$

(b) First output: the **mean** of the two task scores, $\frac{8+4}{2} = 6$. Second output: the **difference** $B - A = 4 - 8 = -4$, i.e. task B is 4 points below task A for this participant. The matrix *is* the recoding rule — row 1 averages, row 2 contrasts — and the second pair of numbers is the one you would actually analyse.

(c) **No.** The rows are $(\frac{1}{2}, \frac{1}{2})$, of norm $\sqrt{\frac{1}{4} + \frac{1}{4}} = \frac{1}{\sqrt{2}}$, and $(-1,1)$, of norm $\sqrt{2}$. They are orthogonal — $(\frac{1}{2}, \frac{1}{2}) \cdot (-1,1) = -\frac{1}{2} + \frac{1}{2} = 0$ — but not normalised, and they are not even of the *same* length. So $\boldsymbol{M}$ stretches the two directions by different factors and does not preserve lengths. This is deliberate here: "mean" and "difference" are the natural units for the interpretation, and readability was preferred to orthonormality.

(d) Consequently $\boldsymbol{M}^\top$ is **not** the way back. Explicitly,

$$\boldsymbol{M}^\top \boldsymbol{M} = \begin{pmatrix} \frac{1}{2} & -1 \\ \frac{1}{2} & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{4}+1 & \frac{1}{4}-1 \\ \frac{1}{4}-1 & \frac{1}{4}+1 \end{pmatrix} = \begin{pmatrix} \frac{5}{4} & -\frac{3}{4} \\ -\frac{3}{4} & \frac{5}{4} \end{pmatrix} \neq \boldsymbol{I}.$$

The genuine inverse must be computed. For a 2×2 matrix, $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)^{-1} = \frac{1}{ad-bc}\left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right)$ (swap the diagonal, flip the signs off it, divide by the determinant). Here $\det \boldsymbol{M} = \frac{1}{2} \cdot 1 - \frac{1}{2} \cdot (-1) = \frac{1}{2} + \frac{1}{2} = 1$, so

$$\boldsymbol{M}^{-1} = \begin{pmatrix} 1 & -\frac{1}{2} \\ 1 & \frac{1}{2} \end{pmatrix}.$$

Check on $(6, -4)$*:*

$$\boldsymbol{M}^{-1} \begin{pmatrix} 6 \\ -4 \end{pmatrix} = \begin{pmatrix} 6 - \frac{1}{2}(-4) \\ 6 + \frac{1}{2}(-4) \end{pmatrix} = \begin{pmatrix} 6+2 \\ 6-2 \end{pmatrix} = \begin{pmatrix} 8 \\ 4 \end{pmatrix} = \boldsymbol{s}. \checkmark$$

In words this is obvious once written out: from the mean m and the difference d you recover $A = m - \frac{d}{2}$ and $B = m + \frac{d}{2}$.

Common mistake. Concluding "the rows are orthogonal, so $\boldsymbol{M}^\top = \boldsymbol{M}^{-1}$". It is *normalisation* that makes the transpose do the work of the inverse. Compare Exercise 21(a), where $\boldsymbol{B}^\top \boldsymbol{B} = \boldsymbol{I}$ held precisely because the columns had norm 1.

Exercise 27 $\boldsymbol{X}$ is the 5×3 table, $\boldsymbol{w} = (1, 2, -1)$: dimensions of $\boldsymbol{X}$, $\boldsymbol{w}$, $\boldsymbol{Xw}$; compute $\boldsymbol{Xw}$; what are X_{42} and X_{24}?

What this exercise uses. Two conventions and one payoff. **Convention 1:** X_{ij} means row i, column j — here participant i, measure j. Rows are observations, columns are variables. **Convention 2:** a product $\boldsymbol{Xw}$ requires the inner dimensions to match: $(5 \times 3) \cdot (3 \times 1)$ is legal and gives 5×1; the two 3's must agree and they disappear. **The payoff:** each entry of $\boldsymbol{Xw}$ is one row of $\boldsymbol{X}$ dotted with $\boldsymbol{w}$, i.e. one participant's score. So a single product applies one scoring rule to everybody at once — and this is the linear part of any regression prediction.

Solution.

(a) $\boldsymbol{X}$ is 5×3 (five participants, three measures); $\boldsymbol{w}$ is a column of three numbers, so 3×1; and the product is $(5 \times 3) \cdot (3 \times 1) = 5 \times 1$ — one number per participant, as it must be.

(b) Row by row — these are the five computations of Exercise 9(a), now performed in one product:

$$\boldsymbol{Xw} = \begin{pmatrix} 4 & 9 & 1 \\ 6 & 7 & 3 \\ 5 & 8 & 2 \\ 9 & 4 & 6 \\ 7 & 6 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 4+18-1 \\ 6+14-3 \\ 5+16-2 \\ 9+8-6 \\ 7+12-4 \end{pmatrix} = \begin{pmatrix} 21 \\ 17 \\ 19 \\ 11 \\ 15 \end{pmatrix},$$

exactly the scores found before. ✓

(c) X_{42} is row 4, column 2: participant 4's accuracy, which is **4**. X_{24} **does not exist** — it would be column 4, and there are only three measures. The order of the indices is a rule, not a stylistic preference: swapping them here asks for something that is not in the table.

Common mistake. Transposing the table mentally and computing $\boldsymbol{w}^\top \boldsymbol{X}$ or $\boldsymbol{X}^\top \boldsymbol{w}$ "because the numbers fit". $\boldsymbol{X}^\top \boldsymbol{w}$ is illegal here (3×5 against 3×1), and when a dimension check fails the fix is to ask which object you actually want, not to flip things until they multiply.

Exercise 28 (a) $\left(\begin{smallmatrix}1&2\\3&4\end{smallmatrix}\right)\left(\begin{smallmatrix}5\\-1\end{smallmatrix}\right)$ by rows and by columns; (b) and (c) two products in both orders; (d) a pair of matrices that do commute.

What this exercise uses. To multiply two matrices, entry (i, j) of $\boldsymbol{AB}$ is (row i of $\boldsymbol{A}$)·(column j of $\boldsymbol{B}$). Composition of transformations is what this computes, and in $\boldsymbol{AB}$ the **right-hand matrix acts first** — it is the one standing next to the vector in $\boldsymbol{A}(\boldsymbol{Bx})$. Since "rotate then stretch" is visibly not "stretch then rotate", matrix multiplication is *not* commutative: $\boldsymbol{AB} \neq \boldsymbol{BA}$ in general. It is, however, associative, and $(\boldsymbol{AB})^\top = \boldsymbol{B}^\top \boldsymbol{A}^\top$ reverses the order too.

Solution.

(a) *By rows:*

$$\begin{pmatrix}1&2\\3&4\end{pmatrix}\begin{pmatrix}5\\-1\end{pmatrix} = \begin{pmatrix}1\cdot 5+2\cdot(-1)\\3\cdot 5+4\cdot(-1)\end{pmatrix} = \begin{pmatrix}5-2\\15-4\end{pmatrix} = \begin{pmatrix}3\\11\end{pmatrix}.$$

By columns:

$$5\begin{pmatrix}1\\3\end{pmatrix} + (-1)\begin{pmatrix}2\\4\end{pmatrix} = \begin{pmatrix}5\\15\end{pmatrix} - \begin{pmatrix}2\\4\end{pmatrix} = \begin{pmatrix}3\\11\end{pmatrix}.\checkmark$$

Same answer, as it must be: the two readings are one computation.

(b)

$$\begin{pmatrix}1&2\\0&1\end{pmatrix}\begin{pmatrix}1&0\\3&1\end{pmatrix} = \begin{pmatrix}1\cdot 1+2\cdot 3 & 1\cdot 0+2\cdot 1\\0\cdot 1+1\cdot 3 & 0\cdot 0+1\cdot 1\end{pmatrix} = \begin{pmatrix}7&2\\3&1\end{pmatrix},$$
$$\begin{pmatrix}1&0\\3&1\end{pmatrix}\begin{pmatrix}1&2\\0&1\end{pmatrix} = \begin{pmatrix}1&2\\3&7\end{pmatrix}.$$

The two results differ, so these matrices do not commute: applying the two transformations in the other order is a different transformation.

(c)

$$\begin{pmatrix}2&1\\0&3\end{pmatrix}\begin{pmatrix}1&-1\\2&0\end{pmatrix} = \begin{pmatrix}2+2&-2+0\\0+6&0+0\end{pmatrix} = \begin{pmatrix}4&-2\\6&0\end{pmatrix},$$
$$\begin{pmatrix}1&-1\\2&0\end{pmatrix}\begin{pmatrix}2&1\\0&3\end{pmatrix} = \begin{pmatrix}2-0&1-3\\4+0&2+0\end{pmatrix} = \begin{pmatrix}2&-2\\4&2\end{pmatrix}.$$

Different again — non-commutativity is the rule, not a special case.

(d) Two *diagonal* matrices always commute:

$$\begin{pmatrix}2&0\\0&3\end{pmatrix}\begin{pmatrix}5&0\\0&-1\end{pmatrix} = \begin{pmatrix}10&0\\0&-3\end{pmatrix} = \begin{pmatrix}5&0\\0&-1\end{pmatrix}\begin{pmatrix}2&0\\0&3\end{pmatrix}.$$

The reason is geometric: both stretch the *same* two axes, and the order in which independent stretches are applied does not matter. Other valid answers: any matrix with a multiple of the identity; $\boldsymbol{A}$ with $\boldsymbol{A}^2$ (or any two powers of the same matrix); $\boldsymbol{A}$ with $\boldsymbol{A}^{-1}$; two rotations of the plane.

Remark. The useful version of this fact: whenever two matrices do *commute, it is because they share some structure — the same axes, or the same underlying transformation. Session 2 turns that observation into the theory of eigenvectors.*

Exercise 29 (a) Transpose $\left(\begin{smallmatrix}1&2\\3&4\end{smallmatrix}\right)$ and $\left(\begin{smallmatrix}1&2&3\\4&5&6\end{smallmatrix}\right)$; (b) verify $(AB)^\top = B^\top A^\top$; (c) dimensions of $\boldsymbol{X}^\top\boldsymbol{X}$ and $\boldsymbol{X}\boldsymbol{X}^\top$.

What this exercise uses. The **transpose** $\boldsymbol{A}^\top$ turns rows into columns: $(\boldsymbol{A}^\top)_{ij} = A_{ji}$. An $m\times n$ matrix transposes into an $n\times m$ one. The rule for products *reverses* the order, $(\boldsymbol{AB})^\top = \boldsymbol{B}^\top\boldsymbol{A}^\top$ — which is forced by the dimensions alone: if $\boldsymbol{A}$ is $m\times n$ and $\boldsymbol{B}$ is $n\times p$, then $\boldsymbol{B}^\top\boldsymbol{A}^\top$ is $(p\times n)(n\times m)$, legal, whereas $\boldsymbol{A}^\top\boldsymbol{B}^\top$ is $(n\times m)(p\times n)$, which does not even multiply unless $m = p$.

Solution.

(a) Rows become columns:

$$\begin{pmatrix}1&2\\3&4\end{pmatrix}^\top = \begin{pmatrix}1&3\\2&4\end{pmatrix}, \qquad \begin{pmatrix}1&2&3\\4&5&6\end{pmatrix}^\top = \begin{pmatrix}1&4\\2&5\\3&6\end{pmatrix}.$$

The second is 2×3 and becomes 3×2.

(b) With $A = \left(\begin{smallmatrix}2&0\\0&3\end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$:

$$AB = \begin{pmatrix}2\cdot 0+0\cdot 1 & 2\cdot 1+0\cdot 0\\ 0\cdot 0+3\cdot 1 & 0\cdot 1+3\cdot 0\end{pmatrix} = \begin{pmatrix}0&2\\3&0\end{pmatrix}, \qquad (AB)^\top = \begin{pmatrix}0&3\\2&0\end{pmatrix}.$$

Both A and B happen to be symmetric here, so $A^\top = A$ and $B^\top = B$, and

$$B^\top A^\top = \begin{pmatrix}0&1\\1&0\end{pmatrix}\begin{pmatrix}2&0\\0&3\end{pmatrix} = \begin{pmatrix}0&3\\2&0\end{pmatrix} = (AB)^\top.\checkmark$$

And to see that the reversal is essential, compute the wrong one:

$$A^\top B^\top = \begin{pmatrix}2&0\\0&3\end{pmatrix}\begin{pmatrix}0&1\\1&0\end{pmatrix} = \begin{pmatrix}0&2\\3&0\end{pmatrix} \neq (AB)^\top.$$

(c) $\boldsymbol{X}$ is 5×3, so $\boldsymbol{X}^\top$ is 3×5 and

$$\boldsymbol{X}^\top\boldsymbol{X}:\ (3\times 5)(5\times 3) = 3\times 3 \quad \text{(measure} \times \text{measure)},$$

$$\boldsymbol{X}\boldsymbol{X}^\top:\ (5\times 3)(3\times 5) = 5\times 5 \quad \text{(participant} \times \text{participant)}.$$

Both products are legal, and they are completely different objects: the first compares measures with each other, the second compares participants.

Remark. $\boldsymbol{X}^\top\boldsymbol{X}$ *is n times the covariance matrix of the measures, once the columns have been centred: its entry (j,k) is the sum over participants of $X_{ij}X_{ik}$, i.e. how measures j and k vary together. It is the object the whole of Session 2 revolves around — rank, eigenvectors, PCA.*